

INFRASTRUCTURE PORTFOLIO

HomeLab

DevOps Infrastructure & AI/ML Experimentation Platform

3-Node Proxmox Cluster

HashiCorp Nomad Orchestration

Hybrid AI/ML Infrastructure

PROJECT OVERVIEW

MISSION

Build a production-grade infrastructure platform for learning modern DevOps practices, container orchestration, and AI/ML experimentation through hands-on implementation.

APPROACH

Systematic, phase-based development

Operational maturity before automation

Documentation-first methodology

FOCUS AREAS

Container orchestration with Nomad

Service discovery and observability

AI/ML workload deployment

VALUES

Practical over perfect

Learn by doing

Incremental progress

Foundation for continuous learning and experimentation

INFRASTRUCTURE ARCHITECTURE

COMPUTE LAYER

3x Proxmox Hosts

Intel-based bare metal servers

Virtual Machine Cluster

Flexible workload distribution

Mac Studio (M1 Max)

GPU-accelerated AI inference

ORCHESTRATION

HashiCorp Nomad

Container orchestration

Consul

Service discovery

Traefik

Reverse proxy

STORAGE

Synology NAS (10TB)

NFS Shared Volumes

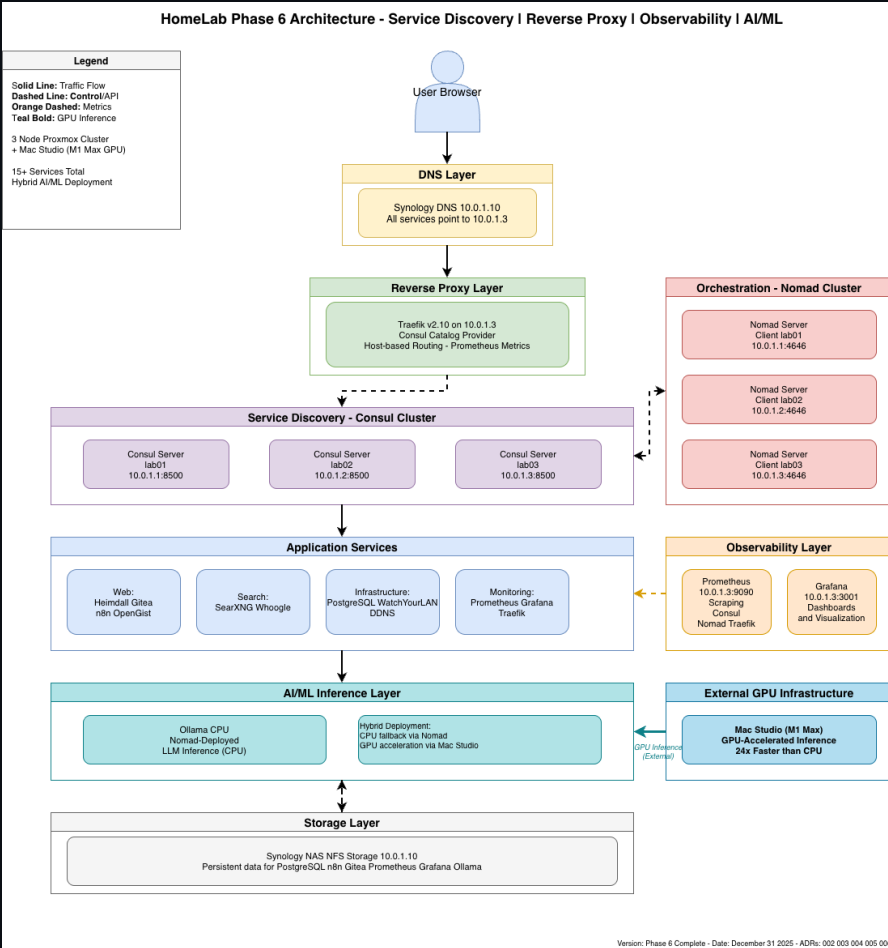
OBSERVABILITY

Prometheus + Grafana

Pushover Alerting

Designed for operational simplicity and learning value

SYSTEM ARCHITECTURE



Layered architecture: DNS → Proxy → Service Discovery → Orchestration → Services → AI/ML → Storage

TECHNOLOGY STACK

CONTAINER PLATFORM

- Docker runtime across all nodes
- LinuxServer.io image ecosystem
- Privileged mode for complex services

SERVICE MESH

- Consul service registry
- Automated health checks
- Dynamic service discovery

AI/ML INFRASTRUCTURE

- Ollama for local LLM inference
- Hybrid GPU/CPU deployment
- 24x performance on M1 Max GPU

MONITORING & ALERTING

- Prometheus metrics scraping
- Custom + community dashboards
- Real-time mobile notifications

Selected for operational simplicity and production-readiness

IMPLEMENTATION JOURNEY

PHASE 1-2 ✓

FOUNDATION

- Hardware inventory documented
- Nomad cluster established
- Architecture decisions recorded

PHASE 3-4 ✓

CORE SERVICES

- Consul service discovery
- Traefik reverse proxy
- Container deployment patterns

PHASE 5 ✓

OBSERVABILITY

- Prometheus metrics stack
- Grafana dashboards
- Mobile alerting system

PHASE 6 ✓

AI/ML PLATFORM

- Hybrid Ollama deployment
- GPU-accelerated inference
- Orchestrated CPU fallback

Progressive buildout with validation at each milestone

THE HYBRID DECISION

Architecture Question: Where Should Ollama Run?

Core Principle

"Learn by doing (CPU), perform in production (GPU)"

Learning Achieved

Nomad AI orchestration mastered

Performance Delivered

24x speedup when needed

Resilience Bonus

Built-in redundancy

Decision: Both endpoints operational, choose based on use case

KEY LEARNINGS

OPERATIONAL MATURITY FIRST

Automate the running system before automating infrastructure provisioning. Manual operations reveal pain points and inform better automation decisions.

PARALLEL ACCESS

Maintain dual access paths during migrations to ensure service availability and enable safe rollback.

DOCUMENTATION TIMING

Document decisions while context is fresh rather than retroactively to capture reasoning and alternatives.

HYBRID ARCHITECTURES

Balance learning value with performance by combining CPU and GPU resources strategically.

FOCUSED SESSIONS

Systematic session-based work produces more reliable results than marathon implementation efforts.

Principles validated through hands-on experience

TECHNICAL ACHIEVEMENTS

24x faster

GPU ACCELERATION

M1 Max GPU vs CPU inference performance

100% uptime

SERVICE AVAILABILITY

Comprehensive health checks and monitoring

3-Node

CLUSTER SCALE

Distributed workload orchestration

PRODUCTION CAPABILITIES

- ✓ Automated service discovery
- ✓ Dynamic load balancing
- ✓ Container health monitoring
- ✓ Mobile alerting integration
- ✓ Custom observability dashboards
- ✓ NFS-backed persistent storage
- ✓ Hybrid AI/ML infrastructure
- ✓ Version-controlled configs
- ✓ Architecture decision records

Enterprise-grade capabilities in a homelab environment

DEPLOYED SERVICES

ORCHESTRATION

Nomad (3 servers, N workers)
Consul service mesh
Traefik reverse proxy

MONITORING

Prometheus metrics
Grafana visualization
Pushover alerting

AI/ML

Ollama (Mac Studio GPU)
Ollama (Nomad CPU)
Local LLM inference

STORAGE

NFS mounts from Synology
Persistent volume support
Shared data access

APPLICATIONS

Containerized workloads
LinuxServer.io images
Custom deployments

NETWORKING

Service-to-service discovery
Automated routing
Health-based failover

All services registered in Consul with comprehensive health checks and observability coverage

Production-ready services with automated management

DEVELOPMENT METHODOLOGY

WORKFLOW PATTERN

Focused work sessions

Real-time documentation

Formal commit process

DOCUMENTATION

Google Drive for strategy

Local Git for configs

Architecture Decision Records

VERSION CONTROL

Git for configurations

Detailed commit messages

DEPLOYMENT

Dry-run validation

Incremental rollout

Systematic approach balancing velocity with reliability

FUTURE ROADMAP

NEXT PHASE: INFRASTRUCTURE AUTOMATION

Automate infrastructure provisioning and configuration management now that operational patterns are proven and well-understood through manual management.

AUTOMATION TARGETS

- Terraform for infrastructure
- Ansible for configuration
- GitOps workflows

AI/ML EXPANSION

- RAG pipeline development
- Agent framework exploration
- Model fine-tuning experiments

CAPABILITIES

- Advanced networking patterns
- Security hardening
- Disaster recovery testing

Continue learning through building - expand capabilities based on real needs

Sustainable growth path with continued focus on practical learning